Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/14/2005
PI Name:	Kandarian, Susan Ph.D.		
Project Title:	Mechanisms of Skeletal Muscle Atrophy Due to Simulated Weightlessness		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
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Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
PI Email:	skandar@bu.edu	**Fax:**	FY 671-353-7567
PI Organization Type:	UNIVERSITY	**Phone:**	617-353-5169
Organization Name:	Boston University		
PI Address 1:	635 Commonwealth Avenue		
PI Address 2:	Rm 443		
PI Web Page:	http://www.bu.edu/		
City:	Boston	**State:**	MA
Zip Code:	02215	**Congressional District:**	8
Comments:			
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No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	1		
COI Name (Institution):	Jackman, Robert (Boston University)		
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It is well known that exposure to spaceflight leads to marked decreases in skeletal muscle size and functional capacity, but the mechanisms underlying these changes are poorly understood. During the previous 3 yrs of NSBRI support, our lab has completed an extensive microarray analysis of soleus muscles over a time course of unloading. These data will be used as a source of candidate genes to test for regulatory roles in the atrophy process using genetic approaches in muscle cell culture and in whole muscle. The overall aim of this project is to identify genes involved in the regulation of protein loss during muscle atrophy. Aim 1: First we will test if candidate genes are sufficient to induce myotube atrophy in cell culture. Initial genes to be studied include those that were activated early with unloading, had sustained activation, and are involved with protein degradation. These genes include cathepsins C, D, and L, the ubiquitin ligases Nedd4 and atrogin-1, and calpastatin. As atrophy also involves decreased protein synthesis, we will also study IGFBP5,

Task Description:	a gene involved in the negative growth control of IGF-1. We are using doxycycline inducible and adenoviral expression systems to limit gene overexpression to differentiated myotubes. As endpoint markers of atrophy we will measure cell diameter and DNA/protein ratio, but we will also measure changes that represent intermediate steps in the atrophy process. These include the phosphorylation state of proteins involved in translation, the activity of proteolytic systems, and protein synthetic and degradation rates. In addition, co-expression strategies will be used to test whether multiple genes more effectively induce markers of atrophy. Where possible we will also test whether the inhibition of candidate genes can attenuate atrophy in a myotube atrophy model, thus testing whether the candidate is necessary for atrophy in cell culture. Aim 2: To determine whether the overexpression or the inhibition of candidate genes can regulate atrophy in vivo. To test if the genes mentioned above are sufficient to induce markers of atrophy in vivo, we will overexpress candidate genes in rat soleus muscle using plasmid electrotransfer, adenovirus or adeno-associated virus infection. To test if a candidate gene is necessary for unloading induced atrophy, we will inhibit its expression by injecting a dominant negative or an anti-sense version of the candidate gene contained in plasmid or viral expression vectors. We will then measure if unloading atrophy can be attenuated in fibers containing the transgene. Uptake by individual muscle fibers will be tracked using epitope-tagged cDNAs or EGFP fusion proteins in order to colocalize overexpression with the effects on fiber size. Knockout mice will also be used to evalute the ability of candidate genes to attenuate unloading muscle atrophy. By identifying genes involved in atrophy we will be in a better position to develop more effective nutritional or pharmacological countermeasures to combat the deleterious changes in muscle function due to exposure to spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	By identifying genes involved in atrophy we will be in a better position to develop more effective nutritional or pharmacological countermeasures to combat the deleterious changes in muscle function due to any type of disuse atrophy.
Task Progress:	In one line of experimentation, Nedd4, a HECT domain containing E3 ubiquitin ligase, was increased over a time course of unloading at both mRNA and protein levels. Nedd4 was found to be localized to the subsarcolemma of muscle cells, consistent with the location of its known substrates. Overexpression of Nedd4 in soleus muscles using electrotransfer of a Nedd4 expression plasmid led to increased ubiquitination of protein from whole muscle homogenates, but it did not affect muscle fiber cross-sectional area. In addition Notch protein, an Nedd4 target was downregulated with 7 days of soleus muscle unloading. Overexpression of a dominant negative Nedd4 plasmid inhibited Nedd4 downregulation by unloading, suggesting that Nedd4 targets the ubiquitination of Notch during unloading. In other experiments on the study of muscle atrophy in cell culture, microarray analysis was performed on mature myotubes in “starved” (non-fed) vs control conditions. A marked change in phenotype was found in the atrophied myotubes but it was quite different from that described in papers on whole muscle atrophy due to unloading, starvation, or cachexia. In a third line of experiments we are constructing adenoviral vectors in order to overexpress genes in muscle cell culture after differentiation has occurred. Adenovial expression vectors also have the advantage of having a high infection efficiency. We have shown so far that adenoviral mediated expression of atrogin-1 in myotubes was preferentially localized to the nucleus in “starved” myotubes but localizes to the cytoplasm in normally fed or in hypertrophied myotubes. In addition, since we showed that an IkB family member called bcl-3 was required for unloading atrophy in vivo, we constructed a fusion protein of EGFP-Bcl-3 and made an adenovirus that expresses this gene. When myotubes are infected with this fusion protein is not only localizes to the nucleus, but it leads to a reorganization of both cell shape and nuclear arrangement. Ongoing experiments will test whether components of the cytoskeleton are involved in this phenotype and whether it is associated with markers of atrophy.
Bibliography Type:	Description: (Last Updated: 09/05/2007)
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Articles in Peer-reviewed Journals	Jackman, R.W. and S.C Kandarian. "The Molecular Basis of Skeletal muscle atrophy." Am J. Physiol. Cell Physiol. 287: C834-C843, , Oct-2004
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